

Run 06 | NME

India

IV only

Target Necessity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 1,849 (6.71%) **Case time** 55s
Train / val. / test 18,740 / 3,305 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: XGBoost (0.9882). Highest F1: CatBoost (0.7807). The all-negative accuracy baseline is 93.29% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
XGBoost	0.9882	0.8227	96.93	75.13	81.03	0.7797	<1s
H2O GBM	0.9881	0.8233	96.86	75.13	79.40	0.7721	4s
CatBoost	0.9880	0.8241	96.95	75.31	81.03	0.7807	9s
Random Forest	0.9869	0.8124	96.88	75.32	79.40	0.7731	1s
Elastic Net	0.9865	0.8030	96.62	73.16	78.32	0.7565	13s
Lasso	0.9864	0.8027	96.64	73.23	78.59	0.7582	10s
Logistic Regression	0.9861	0.7963	95.99	65.04	86.72	0.7433	1s
Ridge	0.9846	0.7791	96.26	71.06	74.53	0.7275	8s
Decision Tree	0.9846	0.7446	96.66	73.07	79.40	0.7610	<1s
Naive Bayes	0.8779	0.4801	93.63	52.24	56.91	0.5447	6s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	679.38	50.12
2	Established business owner	250.33	18.47
3	Knowing entrepreneurs	37.01	2.73
4	Self-efficacy	20.09	1.48
5	Good career choice	18.70	1.38
6	Business exit	18.32	1.35
7	Business angel	16.69	1.23
8	Fear of failure	15.17	1.12

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Investment environment	14.29	1.05
2	Prosperity index score	14.13	1.04
3	Personal freedom	12.58	0.93

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 06_obj2_india_iv_NME. Scores and importance are taken directly from saved CSV outputs.